

Hardware Sensor Test Documentation: Proximity Sensor

Module: Proximity Sensor Testing

Functionality: In-Call Display Control (Ear Detection)

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_PROX_001	Verify screen turning OFF during an active voice call.	Device is in an active carrier or VoIP call. Screen is currently ON.	1. Bring the device close to the ear/cover the top bezel area with an object (within 0-5 cm).	The proximity sensor must detect the object state as "Near", and the screen backlight must turn OFF immediately to prevent accidental face touches.	Pass
TC_PROX_002	Verify screen turning ON when moving away from the ear.	Device is in an active call. Screen is turned OFF by the sensor.	1. Move the device away from the ear/remove the object covering the top bezel.	The sensor must change state to "Far", and the display must instantly turn ON, restoring access to the in-call UI.	Pass
TC_PROX_003	Verify sensor behavior during hands-free (Speakerphone) mode.	Device is in an active call. Speakerphone mode is toggled ON.	1. Cover the top proximity sensor area with a hand during the call.	The display must remain ON and active, ignoring the proximity data since hands-free mode is enabled.	Pass

Functionality: Pocket Mode & Accidental Touch Prevention

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_PROX_004	Verify Pocket Mode triggering when the sensor is fully covered.	Pocket Mode/Pocket Anti-mistouch feature is enabled in Settings.	1. Lock the device. 2. Slide the device into a pocket or cover the top sensor completely. 3. Press the power button to wake the screen.	The lock screen must show a "Pocket Mode Active" warning banner, and all touch inputs on the display must be temporarily blocked.	Pass
TC_PROX_005	Verify Pocket Mode dismissal upon uncovering the sensor.	Pocket Mode UI is currently active on the screen.	1. Pull the device out of the pocket/uncover the top bezel completely.	The warning banner must disappear instantly, and the touch screen must seamlessly return to its normal interactive state.	Pass

Functionality: Hardware Validation & Smart Audio Gestures

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_PROX_006	Verify binary state switching in Factory Diagnostic Tool.	Engineering/Diagnostic Menu (*#899# or *#*#6484#*#*) is open.	1. Leave the sensor uncovered. 2. Cover the sensor area with an opaque card.	The raw value must binary-switch precisely (e.g., from 5.0 cm/Far to 0.0 cm/Near) without bouncing between states.	Pass
TC_PROX_007	Verify smart audio switching in messaging apps (e.g., WhatsApp).	A voice note/audio message is playing through the main loud speaker.	1. Bring the phone close to the ear while the audio is playing.	The messaging app must instantly route the audio output from the bottom speaker to the earpiece receiver at a lower volume.	Pass